

THE ANTICANCER EFFECT OF RAC1 PROTEIN AND NAI COMPOUND IN MHCC97L HEPATOCELLULAR CANCER CELLS



**HKU
Med**

Wen Ka Ning
LKS Faculty of Medicine
School of Biomedical Science
The University of Hong Kong
Mentored by: Amy, Chloe, Neo and Clara
Supervisor: professor Yik Sham Clive Chung
Summer internship program

Abstract

This intern project aims to investigate the anticancer effect of N-acryloylindole (NAI) compound and its capability to target Rac1 in MHCC97L hepatocellular cancer cells using gel-based ABPP and MTT assays.

Introduction

- Rac1 protein belongs to the Rho family of GTPase, playing a crucial role in the recognition of the actin cytoskeleton, responsible for cell motility.
- RHO GTPases participate in the proliferation, migration, invasion, and metastasis of HCC.
- Rac1 inhibition could block sorafenib resistance in HCC by decreasing glycolysis.
- The NAI compound acts as a warhead in this experiment. It can covalently bind to the cysteine residue of the Rac1 protein.
- The covalent modification of Rac1 by NAI can interfere with the guanine nucleotide exchange process, preventing the activation of Rac1.

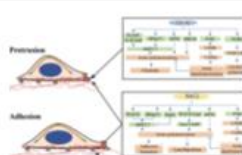


Image 1: RHO GTPase-driven cell migration modes. Rac1 induces the formation of actin-rich lamellipodia and control new protrusions adhere to the ECM through the formation of focal complexes.

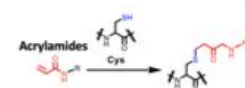


Image 2: Acrylamides can effectively bind to the selenine residue (thiol groups) of proteins

Methodology

Cloning, expression and purification of Rac1

Chemical synthesis of N-acryloylindole (NAI) compound

Gel-based ABPP experiment

MTT assay

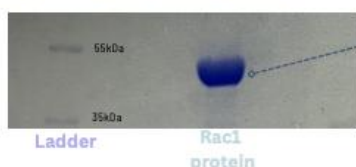
including cloning for GST-Rac1 plasmid, bacteria expression, purification of the Rac1 by using glutathione beads and checking the purity of the Rac1 by SDS-PAGE.

Rac1 protein binding by indole and indoline compounds, then conduct Gel-base ABPP experiment to investigate binding of NAI compounds with Rac1 protein.

MTT assay on MHCC97L liver cancer cells to investigate anticancer properties of NAI compounds.

Results

Commasie Blue result



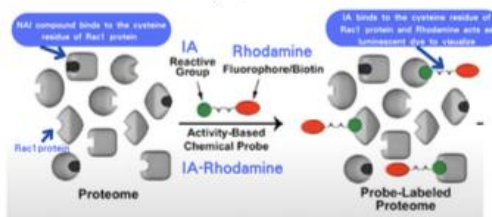
- GST-tagged Rac1 protein
- ~48 kDa

Image 3: The commasie blue result of Rac 1 protein, added 20uL of the 1272.26 mg/ml protein

Gel-based ABPP Result



Image 4: using stain free and IA-Rhodamine to detect the binding of Rac1 protein with NAI compound and Reaction 3 compound (indole and indoline)

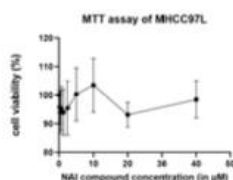


Chemical Result



0.6g 5-hydroxyindole → 0.07g N-acryloylindoline

MTT Result



- The expected result is that cell viability will significantly decrease after adding different concentrations of NAI compounds.

Image 5: The MTT assay result of MHCC97L liver cancer cell, Y axis is the cell viability, X axis is the concentration of NAI compound, using zero concentration of NAI compound as blank.

Conclusion

- The NAI compound effectively binds to and inhibits the Rac1 protein in the HCC, as demonstrated by ABPP method.
- The anticancer effect of the NAI compound on the proliferation of MHCC97L cancer cells remains uncertain, as indicated by the results of the MTT assay.
- The ability of the NAI compound to disrupt Rac1 signaling suggests its potential as a therapeutic agent for targeting Rac1 protein in HCC. By inhibiting Rac1, the NAI compound may suppress tumor cell migration, invasion, and metastasis, which are critical hallmarks of aggressive HCC.

References & Acknowledgement

- [1] Wang, T., Rao, D., Yu, C., Sheng, J., Luo, Y., Xia, L., & Huang, W. (2022). RHO GTPase family in hepatocellular carcinoma. *Experimental Hematology & Oncology*, 11(1), 91. <https://doi.org/10.1186/s40164-022-00344-4>
- [2] Ren, Y.-X., Li, X.-B., Liu, W., Yang, X.-G., Liu, X., & Luo, Y. (2022). The Mechanism of Rac1 in Regulating HCC Cell Glycolysis Which Provides Underlying Therapeutic Target for HCC Therapy. *Journal of Oncology*, 2022, 7319641. <https://doi.org/10.1155/2022/7319641>
- [3] Jaganathan, N., Predescu, D., Zhang, J., Machado, R. F., & Sharma, S. (2016). Rac1-mediated cytoskeleton rearrangements induced by intersectin-1s deficiency promotes lung cancer cell proliferation, migration and metastasis. *Molecular Cancer*, 15(1), 59. <https://doi.org/10.1186/s12943-016-0543-1>

Acknowledgements:

I gratefully acknowledge the support and guidance of the professor Clive Y.S. Chung and mentors Amy, Neo, Chloe, and Clara. The project was supported by LKS Faculty of Medicine, School of Biomedical Science.